HOSPITALITY REVENUE MANAGEMENT, GUEST EXPERIENCE AND HOTEL OPERATIONS ANALYTICS PLATFORM

Comprehensive Business and Technical Project Report

Document Information	Details
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Organisation	LTM
Training programme	Capstone Project Programme
Document version	Final structured review version
Date	09 November 2026

Table of Contents

Sr. No.	Section Title
1	Executive Summary
2	Objectives
3	End-to-End Architecture
4	IDMC Mapping
5	Data Cleansing
6	Bronze Layer
7	Silver Layer
8	Gold Layer
9	Fact Constellation Schema and Description
10	Queries and Techniques
11	Credit Analysis
12	KPI Framework
13	Cortex Analyst
14	Unique Selling Proposition (USP)
15	Streamlit Application
16	Technical Flow – End to End
17	Security and Governance
18	Masking and Error Handling
19	Edge Cases
20	Challenges
21	Burn-down Chart
22	Kanban Chart
23	Conclusion and Future Enhancements

1. Executive Summary

This project delivers a unified hospitality analytics platform that converts fragmented operational data into business-ready insight. Seven source datasets covering reservations, rooms, hotels, guests, billing transactions, feedback and booking events are ingested from AWS S3, profiled and cleansed in Informatica IDMC, and processed through Snowflake Bronze, Silver and Gold layers. Snowpark Python, Snowflake Cortex capabilities and Streamlit extend the warehouse into advanced analytics and user-facing decision support.

The platform addresses inconsistent files, missing values, duplicates, weak historical traceability and disconnected reporting. The technical design preserves the raw source, creates trusted canonical entities, maintains reservation history using SCD Type 2, builds a fact constellation model, secures sensitive data with role-based access and masking, and exposes approved metrics through analytical marts and Streamlit applications.

Headline outcome	**Reported value**
Source records processed	7,160
Records approved by IDMC	7,157 (99.96%)
Room revenue	Approximately INR 4.89 crore
Average daily rate	INR 5,811.20
Room nights sold	8,418
Reservations	2,195
Average guest rating	4.14 out of 5
Recommendation rate	80.6%

The design is suitable for an SME technical review. Production readiness remains dependent on reconciliation of the occupancy denominator, booking-funnel sequence and cross-mart KPI differences. These items are treated as controlled validation issues rather than hidden limitations.

2. Objectives

The objective is to create a centralised, quality-controlled and scalable hospitality analytics solution that converts source data into reliable management information. The report uses the same direct, outcome-oriented style as the project presentation: ingest and validate, process and model, analyse and report.

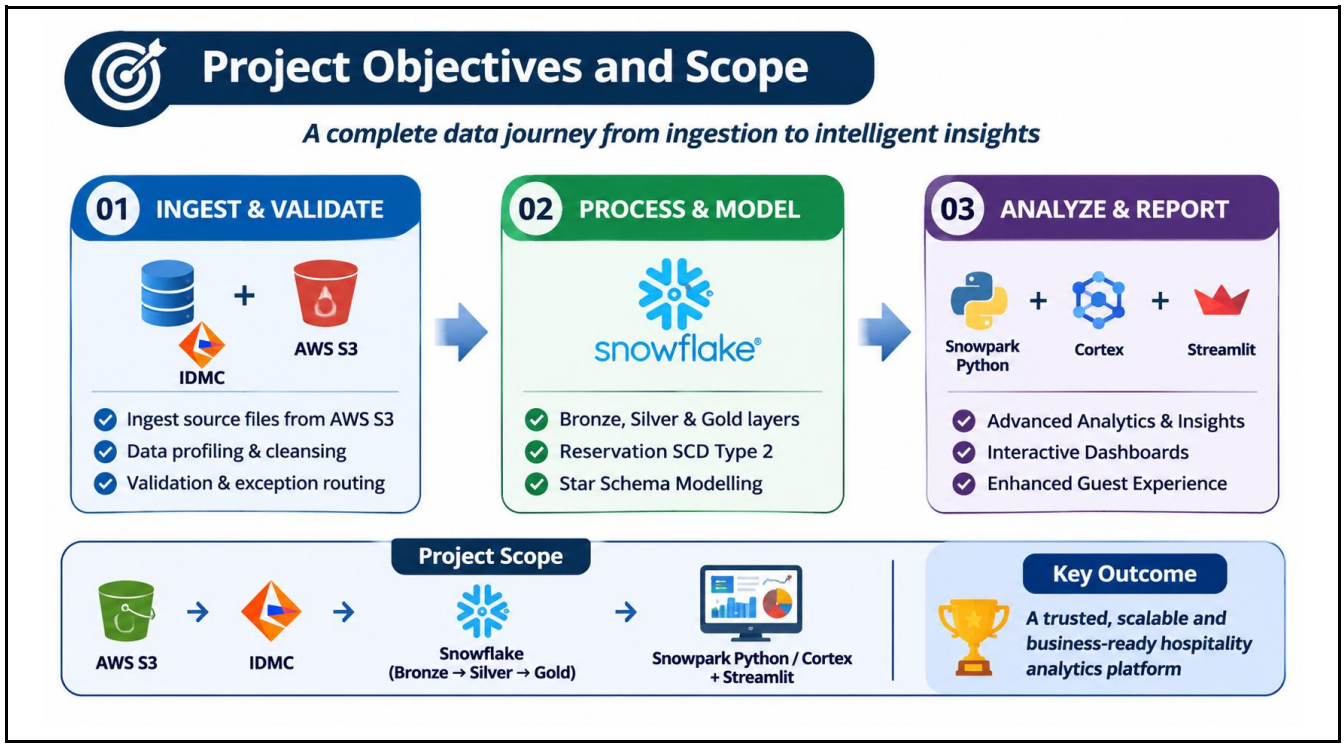


Figure 1. Project objectives and scope from the original presentation.

- **Ingest and validate:** Implementation response – Load CSV and JSON sources from AWS S3 and apply IDMC profiling, rules and exception routing; Business outcome – Reduces manual preparation and prevents invalid data from contaminating reports.

- Process and model: Implementation response – Implement Bronze, Silver and Gold layers, SCD Type 2 and dimensional modelling; Business outcome – Creates traceable, historically consistent analytical data.

- Analyse and report: Implementation response – Build KPIs, advanced analytics and Streamlit dashboards; Business outcome – Enables revenue, guest-experience and operational decisions.

- Secure and govern: Implementation response – Use secure views, masking policies and role hierarchy; Business outcome – Protects guest information while enabling approved business access.

Scope

- Six CSV datasets and one JSON booking-event dataset.
- AWS S3 to Informatica IDMC to Snowflake Bronze, Silver and Gold.
- Snowpark Python and Cortex-supported analytical use cases.
- Streamlit dashboards and business-facing applications.
- Data quality, reconciliation, security and project-delivery evidence.

3. End-to-End Architecture

The target architecture follows a controlled source-to-consumption pattern. Source files enter AWS S3, Informatica IDMC profiles and routes the records, Snowflake stores and transforms the data, and Streamlit presents business outputs. Each component has a separate responsibility, which makes issues easier to isolate and correct.

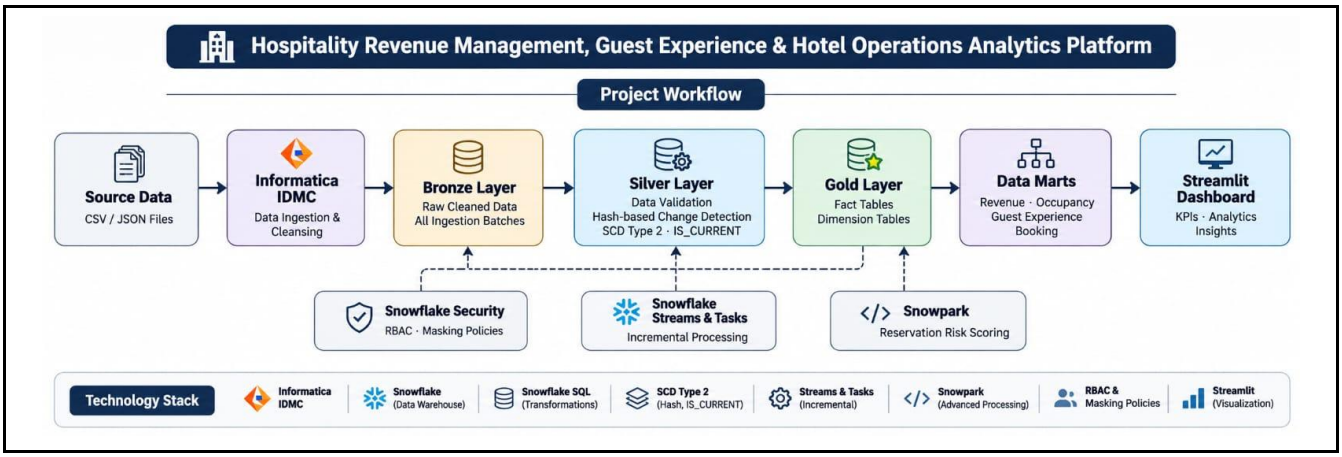


Figure 2. End-to-end architecture extracted from the project PowerPoint.

- Source: Role – AWS S3 stores six CSV files and one JSON file; Key control – File name, path, arrival time and batch registration.
- IDMC: Role – Profiles, standardises, validates, deduplicates and routes records; Key control – Clean, warning, duplicate and reject outputs.
- Bronze: Role – Stores source-aligned records with ingestion metadata; Key control – Replay, lineage and record-count reconciliation.
- Silver: Role – Creates trusted typed and relationship-validated data; Key control – Canonical rules, SCD Type 2 and JSON flattening.
- Gold: Role – Creates conformed dimensions, facts and KPI-ready marts; Key control – Fact grain, metric governance and reconciliation.
- Consumption: Role – Snowpark, Cortex and Streamlit support analytical use cases; Key control – Secure views, masking and role-based access.

4. IDMC Mapping

Informatica IDMC is the data-quality and integration layer. Reusable mappings perform ingestion, sorting, expression-based standardisation, rule evaluation and routing. The project does not treat every row as simply accepted or rejected. Valid records, warning records, duplicates and invalid records follow separate controlled paths.

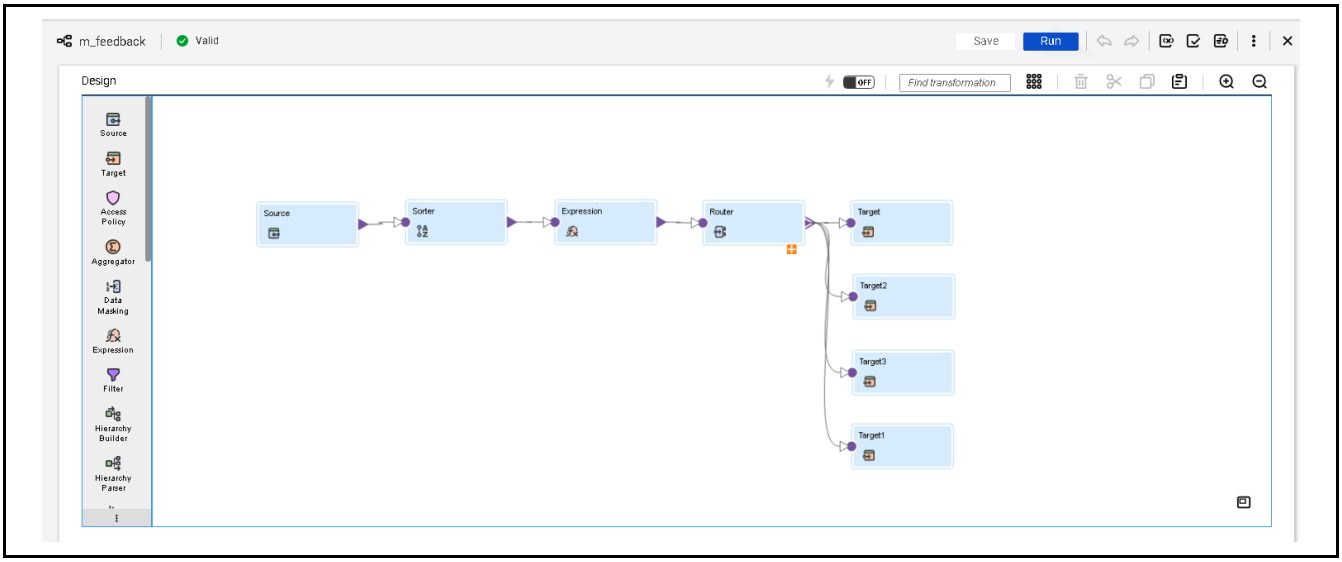


Figure 3. Actual Informatica IDMC mapping screenshot from the implementation.

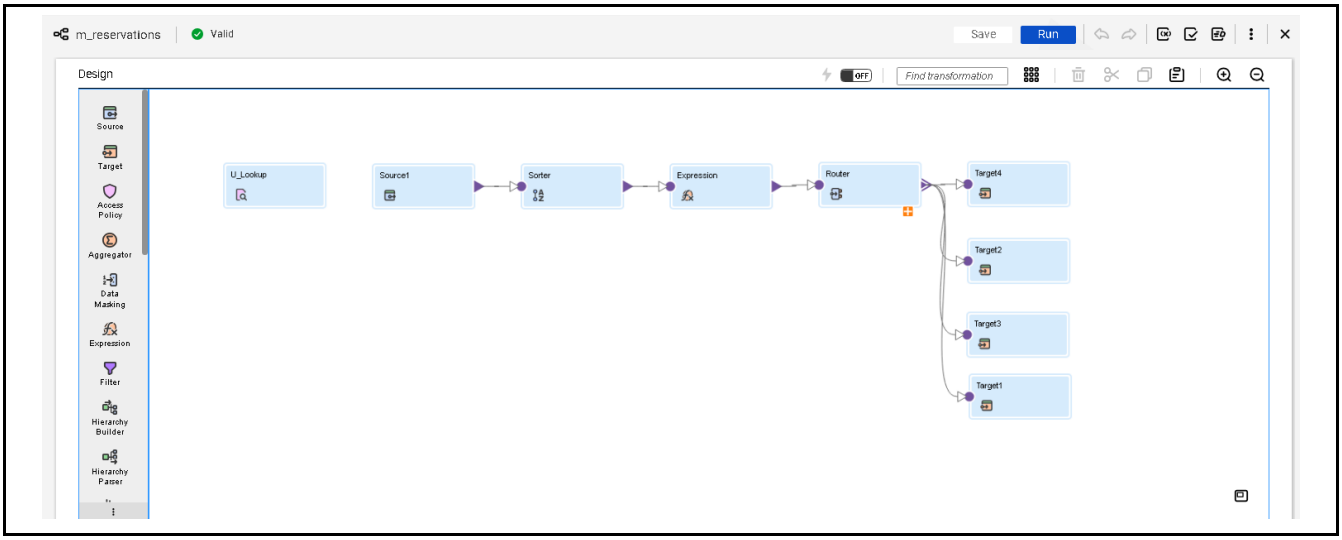


Figure 4. Actual Informatica IDMC transformation and target-routing screenshot.

Mapping sequence

- 1: Mapping action –
  Read source file and attach source metadata; Output – Source-aligned rows.
- 2: Mapping action –
  Standardise blank values, dates, text and numeric fields; Output – Consistent technical representation.
- 3: Mapping action – Apply mandatory-field, format and domain rules; Output – Validation flags and reason codes.
- 4: Mapping action – Detect duplicate business identifiers and repeated rows; Output – Survivor and duplicate routes.
- 5: Mapping action – Validate relationships such as reservation-to-guest and room-to-hotel; Output – Trusted or exception outcome.
- 6: Mapping action – Write clean, warning, duplicate and reject targets; Output – Auditable processing result.

5. Data Cleansing

Data cleansing prepares source records for trusted analytical use. The process preserves the original value for lineage while creating a standardised value for downstream processing. Cleansing is rule-driven and repeatable, not a manual correction performed only for presentation purposes.

Cleansing area	Technique	Example
Missing values	Trim, convert empty strings to null and apply mandatory-field rules.	A missing reservation identifier is rejected; a missing optional description is warned.
Dates and timestamps	Parse accepted formats and standardise to DATE or TIMESTAMP.	Check-out date must not be earlier than check-in date.
Numeric values	Cast to fixed-precision types and apply valid ranges.	Rating must be from 1 to 5.
Text values	Trim spaces and map approved labels.	Status variants are mapped to controlled reservation statuses.
Duplicates	Compare business keys and record hashes.	Repeated reservation rows are separated from valid updates.
Relationships	Validate foreign business keys.	A room must belong to the selected hotel.
Semi-structured data	Validate and flatten JSON.	Booking-event attributes are extracted while retaining the raw payload.

Quality routing

- Clean: all critical validations pass.
- Warning: the record is usable but contains a non-critical issue.
- Duplicate: the record repeats an existing business event or source delivery.
- Reject: the record cannot be promoted without creating an incorrect fact, key or amount.

6. Bronze Layer

The Bronze layer is the first durable Snowflake layer. Bronze stores data close to the source format and preserves the evidence required for replay, reconciliation and defect investigation. Bronze is not intended for management reporting because source inconsistencies may still be present.

Bronze table	Grain	Important metadata
BRONZE_RESERVATIONS	One ingested reservation row	source file, batch ID, row number, ingestion timestamp, record hash
BRONZE_ROOMS	One ingested room row	source file, batch ID, ingestion timestamp
BRONZE_BILLING	One ingested transaction row	source file, batch ID, ingestion timestamp
BRONZE_BOOKING_EVENTS	One JSON event object	raw payload, file, batch, timestamp
BRONZE_GUESTS	One ingested guest row	source file, batch ID, ingestion timestamp
BRONZE_FEEDBACK	One ingested feedback row	source file, batch ID, ingestion timestamp
BRONZE_HOTELS	One ingested hotel row	source file, batch ID, ingestion timestamp

Bronze loading controls

- Register every file before loading.
- Use append-oriented batch processing rather than silent overwrites.
- Identify duplicate deliveries using file or record hashes.
- Store rejection reason and route status where applicable.
- Reconcile source rows to clean, warning, duplicate and rejected outcomes.
- Keep replay and restart behaviour idempotent.

Bronze exit criteria

Every received row must have a known batch and source reference; parse failures must have reproducible error messages; the file and row counts must reconcile; and the accepted set must be available for deterministic Silver processing.

7. Silver Layer

The Silver layer creates trusted canonical data. It standardises types and codes, resolves duplicates, validates relationships, flattens JSON and maintains reservation history. Silver is the operational truth used to build the Gold analytical model.

- Reservations: Standardise dates and statuses, calculate stay nights, validate relationships, deduplicate and maintain SCD Type 2 history.
- Rooms: Standardise room type, bed type, capacity and base price; validate hotel relationship.
- Hotels: Standardise property, geography, hotel type and star rating.
- Guests: Standardise profile and loyalty attributes; isolate sensitive fields.
- Billing: Standardise amount, currency, payment method and status; validate reservation relationship.
- Feedback: Validate rating, recommendation and text; associate feedback with the correct stay.
- Booking events: Flatten JSON, standardise event names and timestamps, remove repeated events and prepare sessions.

SCD Type 2 reservation history

When a tracked reservation attribute changes, the current record is expired and a new current version is inserted. RESERVATION_ID is the natural key; EFFECTIVE_FROM, EFFECTIVE_TO, IS_CURRENT, SOURCE_UPDATED_TIMESTAMP and RECORD_HASH support historical analysis and auditability.

- Current-row uniqueness: Exactly one current version per reservation.
- Date validity: Effective periods do not overlap.
- Change detection: Only tracked attribute changes create a new version.
- Late-arriving updates: Handled by a documented back-dating policy.
- Gold consumption: Facts explicitly use either the current version or the version valid at the analytical date.

8. Gold Layer

The Gold layer is the business-ready analytical layer. Gold resolves Silver business keys to conformed dimensions, loads facts at a declared grain, calculates governed measures and exposes secure marts for Streamlit and advanced analytics.

- Conformed dimensions: Provide one definition of guest, hotel, room and date across analytical subjects.
- Fact tables: Store measurable reservation, billing, feedback and booking-event activity.
- Business marts: Present revenue, occupancy, channel, experience, funnel, risk and pricing views.
- Secure views: Expose only approved rows and columns to each role.
- KPI layer: Centralise formulas and filters so dashboard pages do not calculate differently.

Gold loading sequence

- Load DIM_DATE for the complete reporting horizon.
- Load DIM_HOTEL, DIM_ROOM and DIM_GUEST before dependent facts.
- Resolve natural keys to surrogate keys.
- Use an approved unknown member for monitored late-arriving dimensions.
- Load reservation, billing, feedback and event facts incrementally.
- Reconcile Gold inserts and updates to the eligible Silver change set.

9. Fact Constellation Schema and Description

The project uses a fact constellation, also called a galaxy schema, because multiple fact tables share common dimensions. This design supports cross-domain analysis without repeating inconsistent hotel, guest, room or date definitions.

Object	Declared grain	Description
DIM_GUEST	One approved analytical guest version	Geography, loyalty and permitted profile attributes.
DIM_HOTEL	One hotel	Property, location, hotel type and star rating.
DIM_ROOM	One room analytical entity	Room type, capacity, bed type and base price.
DIM_DATE	One calendar date	Day, week, month, quarter, year and weekend indicators.
FACT_RESERVATION	One reservation at the approved reporting grain	Booking, stay dates, channel, status, room nights and amount.
FACT_BILLING_TRANSACTION	One billing transaction	Amount, currency, payment method and transaction status.
FACT_GUEST_FEEDBACK	One feedback response	Rating, recommendation, comment and sentiment output.
FACT_BOOKING_EVENT	One cleaned booking event	Event stage, timestamp, session, channel and status.

Schema design controls

- Write the fact grain before defining measures.
- Do not mix room-night and reservation-level measures without explicit aggregation logic.
- Retain natural identifiers for traceability and surrogate keys for analytical joins.
- Use role-playing date keys for booking, check-in, check-out, transaction and feedback dates.
- Reconcile every fact to its eligible Silver input.

10. Queries and Techniques

Queries and techniques are designed to be explainable, repeatable and compatible with Snowflake. The following patterns represent the technical approach implemented or recommended for the project.

Technique	Project use	Illustrative logic
CTE-based SQL	Break complex KPI logic into readable stages.	Filter valid reservations, calculate room nights, then aggregate by hotel and month.
MERGE	Load incremental dimension or fact changes.	Match on business key; update current metadata or insert new rows.
ROW_NUMBER	Resolve duplicates using deterministic precedence.	Partition by business key and order by source update and ingestion time.
HASH	Detect record changes efficiently.	Hash tracked attributes and compare incoming with current Silver row.
FLATTEN	Convert booking-event JSON arrays or objects into rows.	LATERAL FLATTEN over the Bronze VARIANT payload.
Window functions	Rank hotels, calculate running totals and compare periods.	RANK, LAG, SUM OVER and moving averages.
Conditional aggregation	Calculate status and channel metrics in one grouped query.	SUM with CASE expressions for confirmed, cancelled or paid events.
Secure views	Expose role-approved analytical columns.	Mask or exclude guest contact information.

Representative query patterns

ADR = SUM(approved room revenue) / NULLIF(SUM(sold room nights), 0).

Occupancy = occupied room nights / available room nights for the same hotel and date scope. Recommendation rate = recommended responses / valid recommendation responses. Funnel conversion = distinct confirmed sessions / distinct eligible sessions after event deduplication and sequence validation.

All production queries should state the reporting date basis, included statuses, currency policy, fact grain and expected reconciliation source. This prevents technically valid queries from producing conflicting management numbers.

11. Credit Analysis

For this hospitality project, credit analysis is interpreted as billing and payment-credit analysis rather than banking credit scoring. The source scope includes billing transactions but does not include borrower income, bureau score, loan exposure or repayment history. Therefore, the project should not claim a lending-risk model.

Billing and credit-status analysis

Analysis area	Business question	Measure
Payment success	How much billed value has been successfully collected?	Successful transaction value and success rate.
Failed payments	Which channels, hotels or methods create payment failure?	Failed count, failed value and failure rate.
Refund and reversal	How much value has been reversed or refunded?	Refund amount, reversal count and ratio to successful payments.
Outstanding exposure	Which confirmed reservations lack a successful linked payment?	Reservation value without an approved transaction.
Payment method mix	Which methods contribute the most collected revenue?	Successful amount by payment method.
Guest credit indicator	Where approved by policy, which stays require additional follow-up?	Rule-based flag using transaction outcome, not a formal credit score.

This section deliberately uses accurate terminology. A future formal credit-risk model would require additional governed features, labelled outcomes, fairness review and validation. The current implementation supports payment and receivables analysis only.

12. KPI Framework

The KPI framework provides one approved definition for each business measure. Every KPI must have a purpose, formula, grain, date basis, filters, owner, refresh schedule and reconciliation source.

KPI	Formula	Primary decision
Room revenue	Approved room-related revenue for valid business statuses.	Measure commercial performance.
ADR	Room revenue divided by sold room nights.	Evaluate realised room rate.
Occupancy rate	Occupied room nights divided by available room nights for the same properties and dates.	Evaluate inventory utilisation.
Recommendation rate	Recommended responses divided by valid responses.	Monitor guest advocacy.
Booking conversion	Distinct confirmed sessions divided by distinct eligible sessions.	Identify funnel effectiveness.
Payment success rate	Successful transactions divided by valid attempted transactions.	Monitor billing performance.
High-risk reservation share	High-risk reservations divided by scored reservations.	Prioritise operational follow-up.

KPI governance

- Use the same numerator and denominator scope.
- Display the reporting period and filter scope on every Streamlit page.
- Treat the near-0.1% occupancy value as under review until its available-room-night denominator is reconciled.
- Do not publish funnel conversion until stage order and session identity are validated.
- Automate source-to-Gold and Gold-to-dashboard reconciliation tests.

13. Cortex Analyst

Cortex Analyst is positioned as a governed future consumption capability over approved Gold-layer semantic data. The objective is to let authorised users ask natural-language questions while keeping metric definitions, relationships and security under enterprise control.

Planned use cases

- Ask revenue questions by hotel, month, room type or booking channel.
- Summarise approved guest-feedback themes.
- Compare ADR and room-night trends across periods.
- Explain changes using governed KPIs instead of generating new formulas.
- Support analysts with query generation over approved semantic objects.

Implementation controls

- Semantic model: Expose business-friendly names, relationships, synonyms and valid measures.
- Security: Apply the same Snowflake roles, secure views and masking policies used by Streamlit.
- Metric governance: Cortex responses must use approved KPI definitions.
- Prompt scope: Restrict access to approved Gold objects, not raw guest data.
- Validation: Test representative questions and compare generated results with certified SQL.
- Transparency: Show source objects, filters and reporting period where practical.

Cortex Analyst should be presented as a planned or partially explored enhancement unless a working semantic model and validated responses are demonstrated during the review.

14. Unique Selling Proposition (USP)

The project USP is the combination of commercial analytics and guest-recovery intelligence on one governed platform. The solution does not stop after displaying ratings. Feedback, reservation context and operational rules are combined to identify the next best action for the hotel team.

Guest Feedback Analysis and Recovery Recommendation Engine

The **Guest Feedback Analysis and Recovery Recommendation Engine** is a key USP of our Hospitality Analytics Platform. The solution goes beyond simply displaying guest ratings and feedback. It analyses guest responses together with reservation, hotel and room context to understand the nature of the guest issue, determine its priority and recommend the **next best recovery action** for the hotel team.

Input	Processing	Business output
Guest rating, service rating and recommendation	Validate, standardise and associate with reservation and property context.	Trusted guest-experience profile.
Feedback text	Classify issues and, where enabled, evaluate sentiment and themes.	Service category and problem summary.
Stay and room context	Connect the issue with hotel, room type and reservation details.	Operational context for investigation.
Priority rules	Apply High, Medium or Low priority based on severity and business conditions.	Actionable recovery queue.
Recommendation logic	Select an explainable action such as guest contact, housekeeping follow-up or room-service escalation.	Next-step recommendation for the team.

The engine converts unstructured guest feedback into **structured and actionable business information**. Instead of requiring hotel teams to manually review every comment, the solution identifies the issue, connects the feedback with relevant stay information and prioritises cases that require attention.

The key differentiator is the **Recovery Recommendation Engine**. After identifying the guest issue and its priority, the platform recommends an appropriate next step, such as contacting the guest, initiating a housekeeping follow-up or escalating a room-service problem. This transforms guest feedback from a passive reporting metric into an **operational decision-support mechanism**.

Through the interactive dashboard, management can analyse **guest ratings, recommendation levels, issue categories, priority levels and individual feedback records**, while the recovery recommendations help operational teams understand where action is required.

From a business perspective, this USP supports **faster identification of dissatisfied guests, structured service recovery, improved operational responsiveness and better guest-experience management**.

Smart Pricing Intelligence & Revenue Opportunity Engine

A second differentiator is the **Pricing Intelligence & Revenue Opportunity Engine**. The solution goes beyond displaying historical revenue and ADR by analysing room-rate patterns, comparing actual reservation prices with relevant historical benchmarks, identifying unusual pricing, and helping hotel managers evaluate potential revenue opportunities.

Input	Processing	Business output
Reservation rate and stay details	Analyse actual nightly rates using historical reservation data.	Trusted reservation-level pricing profile.
Hotel and room context	Compare rates within similar hotel and room-type combinations.	Relevant pricing benchmark.
Historical and seasonal patterns	Establish benchmark rates based on comparable historical bookings and seasonal behaviour.	Expected pricing range for comparison.
Actual vs benchmark rate	Detect unusually high or low prices and calculate pricing variance.	Pricing anomaly and potential revenue-opportunity flag.
Pricing scenario selected by manager	Apply the What-If Revenue Simulator to estimate the financial impact of an alternative room rate.	Estimated revenue impact to support pricing decisions.

The engine presents these findings through an interactive dashboard containing **pricing KPIs, actual-versus-benchmark comparisons, anomaly analysis and reservation-level details**. The key feature is the **What-If Revenue Simulator**, which allows hotel managers to experiment with different room-pricing scenarios and understand their potential revenue impact.

Together, the two USPs complement each other: the **Guest Feedback Analysis and Recovery Recommendation Engine** focuses on improving guest experience and determining the next best recovery action, while the **Pricing Intelligence & Revenue Opportunity Engine** focuses on identifying pricing opportunities and supporting better revenue decisions. Both use the same governed Gold-layer data instead of creating separate analytical silos.

15. Streamlit Application

Streamlit is the business-consumption layer. The application translates the Gold marts into interactive pages for revenue, occupancy, channels, guest experience, booking conversion, reservation risk, guest recovery and pricing intelligence.

Application page	Primary content	Business user
Executive overview	Room revenue, ADR, room nights, reservations, rating, recommendation and peak month.	Hotel manager and SME reviewer.
Revenue and occupancy	Monthly trends, hotel comparison and inventory utilisation.	Revenue analyst.
Booking channels	Channel contribution, booked nights, revenue and ADR.	Commercial and revenue teams.
Guest experience	Ratings, recommendation, trends, heatmap and hotel ranking.	Guest-experience team.
Guest Recovery Assistant	Priority cases, detected issues and recommended actions.	Operations and service teams.
Reservation risk	Low, Medium and High rule-based categories.	Reservation operations.
Pricing intelligence	Benchmark comparison and rate opportunity flags.	Revenue management.

Application design controls

- Display reporting period, active filters and last refresh time.
- Use secure Gold views rather than querying raw Bronze data.
- Label rule-based outputs accurately and do not present them as trained machine learning.
- Show data-quality warnings where a KPI remains under review.
- Ensure every displayed total reconciles to certified SQL.

Dashboard Implementation Screenshots

The following screenshots provide direct implementation evidence of the Streamlit business-consumption layer. Each screenshot is presented separately with a bordered frame, descriptive caption and spacing for clear review.

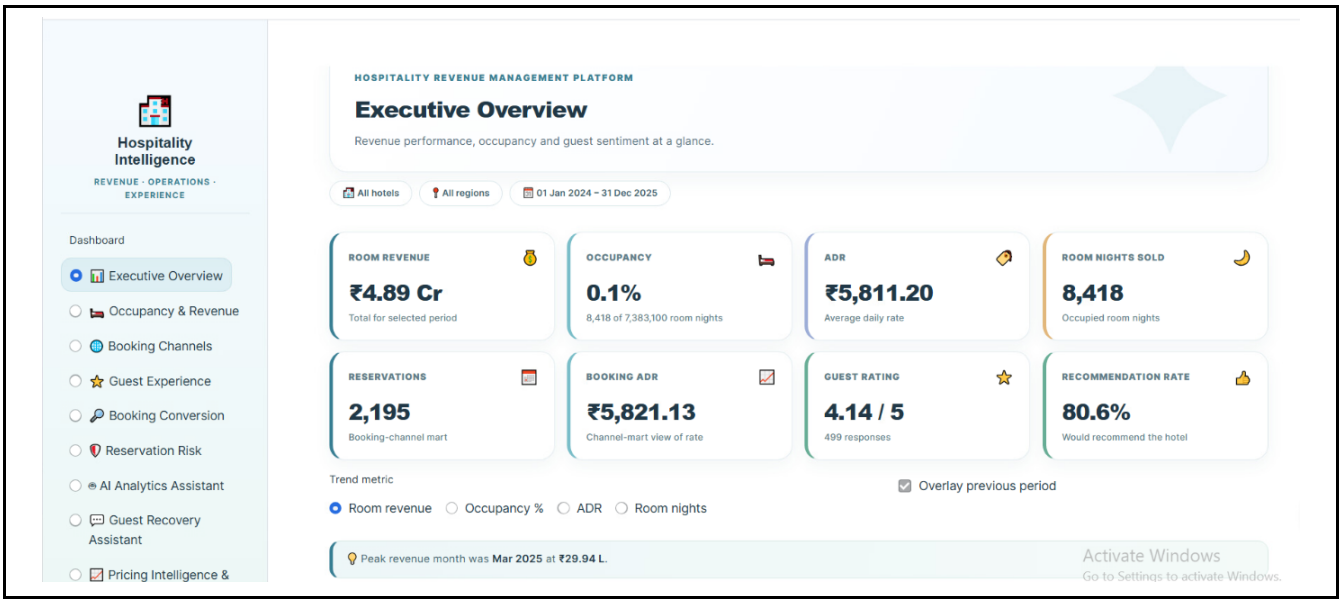


Figure 6. Streamlit Executive Overview displaying room revenue, occupancy, ADR, room nights sold, reservations, booking ADR, guest rating and recommendation rate.

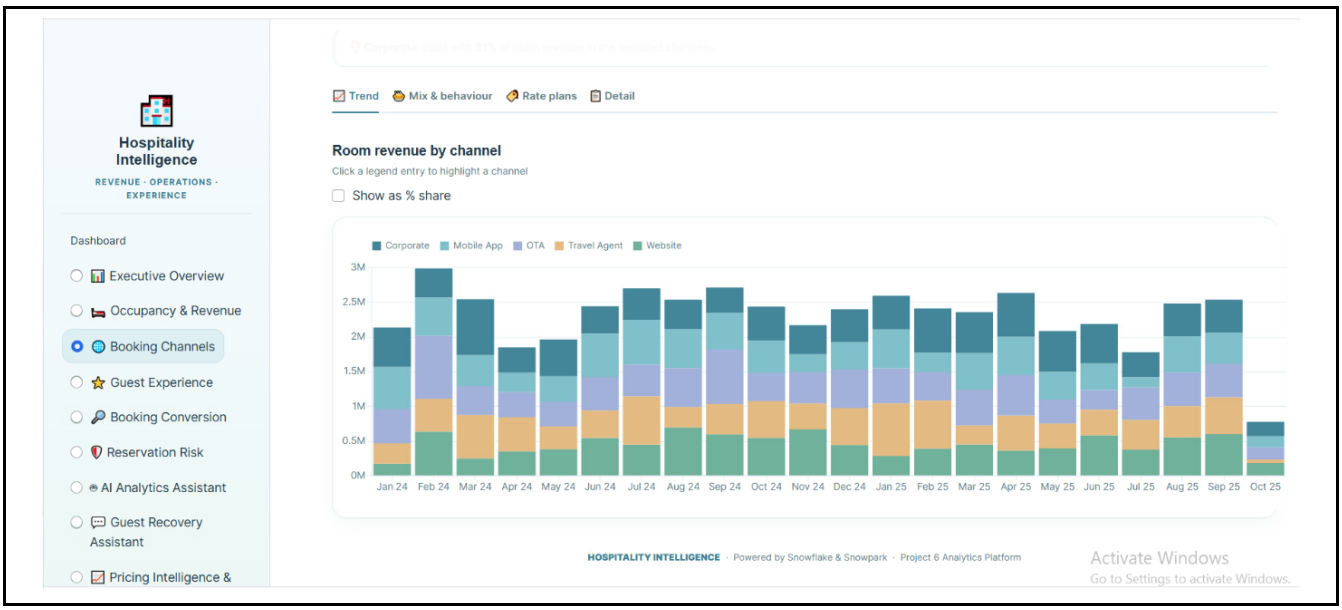


Figure 7. Streamlit Booking Channels dashboard displaying monthly room-revenue contribution by Corporate, Mobile App, OTA, Travel Agent and Website channels.

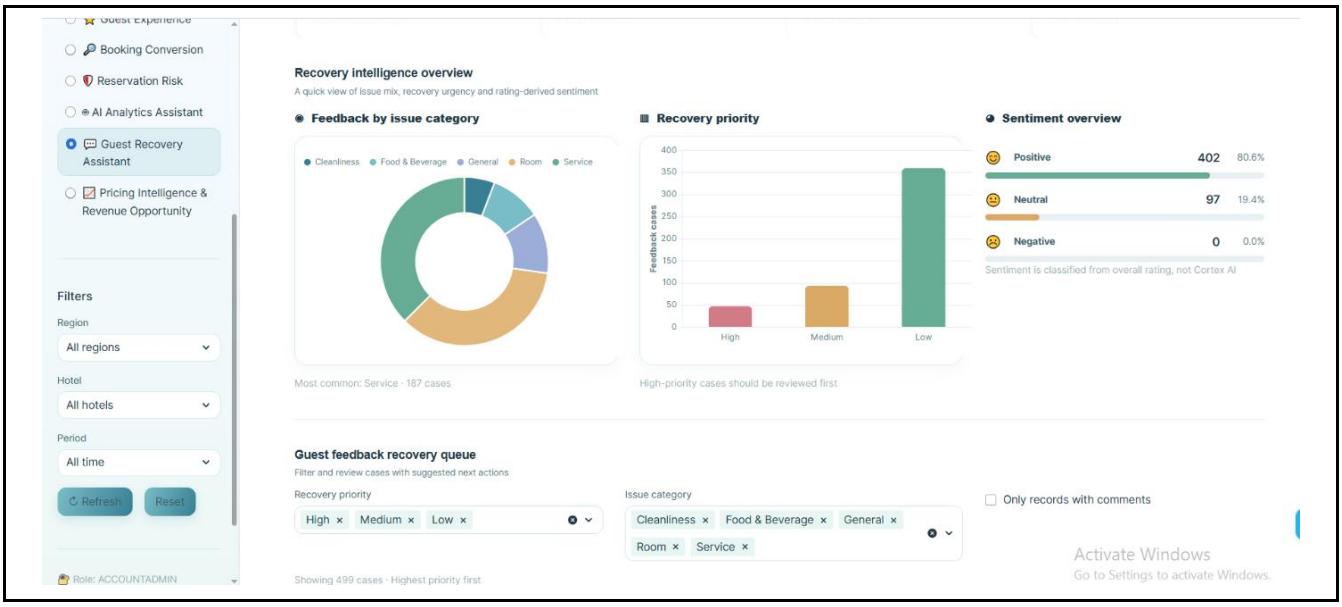


Figure 8. Streamlit Guest Recovery Assistant displaying feedback issue categories, recovery priority, sentiment overview and the guest feedback recovery queue.

16. Technical Flow – End to End

The complete technical flow connects source arrival, quality management, layered transformation, analytical modelling, security and application consumption.

- 1. Source arrival: Technical activity – CSV and JSON files land in AWS S3; Control evidence – File path, name, arrival time and batch registration.
- 2. IDMC profiling: Technical activity – Measure completeness, uniqueness, formats and anomalies; Control evidence – Profile output and quality findings.
- 3. Cleansing and routing: Technical activity – Standardise and separate clean, warning, duplicate and reject rows; Control evidence – Target counts and reason codes.
- 4. Bronze loading: Technical activity – Store source-aligned records and raw JSON; Control evidence – Batch metadata and row-count reconciliation.
- 5. Silver processing: Technical activity – Type, deduplicate, validate relationships, flatten JSON and maintain SCD history; Control evidence – Trusted-row counts, duplicate decisions and SCD tests.
- 6. Gold modelling: Technical activity – Load conformed dimensions, facts and business marts; Control evidence – Surrogate-key resolution and source-to-fact reconciliation.
- 7. Security: Technical activity – Apply role grants, secure views and masking; Control evidence – Positive and negative access tests.
- 8. Analytics: Technical activity – Calculate KPIs, risk rules, recovery recommendations and pricing benchmarks; Control evidence – Certified queries and test cases.
- 9. Streamlit / Cortex: Technical activity – Provide dashboards and governed analytical interaction; Control evidence – App validation and response reconciliation.

Traceability example

A reservation can be traced from its AWS S3 source row through the IDMC clean or exception route, Bronze batch metadata, Silver standardisation and version history, Gold dimension and fact keys, and finally the Streamlit KPI or detailed business page. This end-to-end record walk is recommended for the SME demonstration.

17. Security and Governance

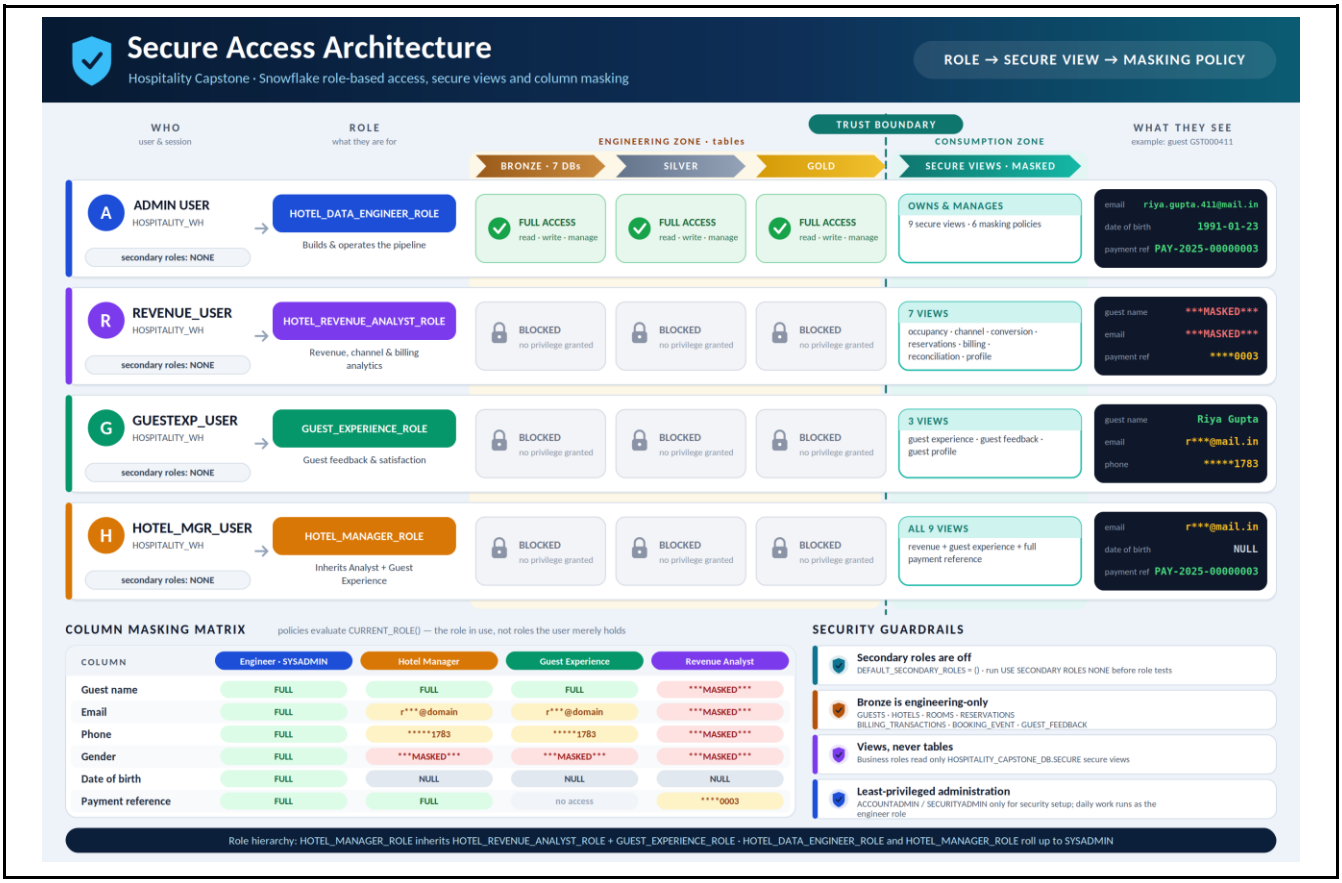


Figure 9. Secure-access architecture from the original presentation.

Role	Approved access
HOTEL_REVENUE_ANALYST_ROLE	Revenue, occupancy, channel, pricing and approved billing measures.
GUEST_EXPERIENCE_ROLE	Ratings, feedback, recovery priority and approved guest context.
HOTEL_MANAGER_ROLE	Consolidated property-level managerial views through inherited roles.
HOTEL_DATA_ENGINEER_ROLE	Pipeline, model and controlled support access.

Governance controls

- Least-privilege grants at database, schema, view and warehouse level.
- Secure views for business consumption.
- Masking policies for sensitive guest attributes.
- Positive tests confirming permitted access.
- Negative tests confirming denied or masked access.
- Audit evidence for role changes and periodic access review.
- Central KPI definitions and source-to-target documentation.

18. Masking and Error Handling

Masking and error handling are combined under one dedicated control section because both protect the reliability and appropriate use of the platform.

Masking

Data type	Recommended masking behaviour	Permitted user
Guest email	Partially or fully mask unless required for approved recovery action.	Authorised guest-experience role.
Guest phone	Mask most digits by default.	Approved operational role only.
Personal identifier	Tokenise or mask in analytical views.	Engineering support under controlled access.
Feedback text	Restrict raw comments when they contain personal information.	Guest-experience role.
Payment reference	Expose only non-sensitive analytical status and method.	Revenue or billing analyst.

Error handling

Error class	Handling approach
File/schema error	Fail or quarantine the batch and retain the structure comparison.
Row parse error	Route the row to reject with the raw value and reason.
Business-rule error	Reject or warn based on severity.
Relationship error	Move to suspense or reject until reference data is available.
Duplicate delivery	Stop repeat loading using file or record hash.
Pipeline failure	Record the failed step, stop downstream promotion and restart idempotently.
Dashboard error	Show a controlled message and do not display partial KPI results as final.

19. Edge Cases

Edge cases are conditions that may be rare but can materially affect hospitality metrics or pipeline correctness. The project should test these explicitly rather than relying only on normal records.

Edge case	Potential impact	Expected handling
Check-out before check-in	Negative room nights and incorrect ADR.	Reject as a critical date error.
Same-day stay	Zero nights if the rule is not defined.	Apply the approved day-use business rule.
Reservation updated after cancellation	Incorrect current status or duplicated history.	Apply ordered SCD Type 2 logic using source update time.
Room belongs to a different hotel	Invalid occupancy and hotel revenue attribution.	Reject or hold until the relationship is corrected.
Refund larger than payment	Negative revenue distortion.	Flag for financial review and exclude from normal recognised revenue.
Confirmed event without booking-started event	Funnel sequence becomes unreliable.	Flag session and exclude from certified funnel until rules are approved.
Duplicate confirmation event	Inflated conversion.	Deduplicate using event ID or stable event signature.
Anonymous guest event	Broken guest join.	Retain session-level analysis without inventing a guest identity.
Missing recommendation response	Incorrect recommendation rate denominator.	Exclude null from valid-response denominator.
Late-arriving dimension	Null fact key or lost record.	Use monitored unknown member and backfill later.
Rerun after partial failure	Duplicate facts.	Use pipeline run IDs, watermarks and idempotent merges.
Very large available-room denominator	Artificially low occupancy.	Reconcile property inventory, date range and active-room logic.

20. Challenges

Challenge	Business impact	Resolution or next action
Fragmented files and formats	Manual effort and inconsistent reporting.	Centralise ingestion and apply reusable IDMC rules.
Nulls, duplicates and invalid values	Distorted facts and KPIs.	Route by quality status and retain reason codes.
Reservation history	Previous states become invisible.	Implement and test SCD Type 2.
Semi-structured booking events	Funnel stages are difficult to calculate reliably.	Preserve raw JSON, flatten in Silver and validate sessions.
Occupancy near 0.1%	Potentially misleading executive conclusion.	Revalidate the available-room-night denominator.
Confirmed sessions exceed started sessions	Conversion metric is not production-ready.	Validate event order, deduplication and session definitions.
Cross-mart total differences	Users may lose confidence.	Create a KPI catalogue and automated reconciliations.
Rule-based risk described as AI	Overstates analytical maturity.	Label the method accurately and document thresholds.
Short project timeline	Reduced test depth.	Use Kanban, burn-down tracking and critical-path testing.

20.1 Testing and Validation Strategy

A structured testing strategy is required to demonstrate that the hospitality analytics platform produces complete, accurate and repeatable results. Testing should cover the full data journey from AWS S3 source files to Informatica IDMC routing, Snowflake Bronze, Silver and Gold processing, and final Streamlit presentation.

The objective is not only to confirm that the pipeline runs successfully, but also to prove that records, relationships and business measures remain correct at every stage.

Testing area	Validation activity	Expected evidence
Source ingestion	Compare received file names, formats and source row counts with batch-registration records.	File inventory, batch ID, arrival timestamp and source count.
IDMC data quality	Test mandatory fields, formats, duplicates, ranges and relationship rules using valid and invalid sample records.	Clean, warning, duplicate and reject counts with reason codes.
Bronze layer	Confirm all accepted source rows are stored with source file, batch and ingestion metadata.	Source-to-Bronze reconciliation and replay evidence.
Silver layer	Validate type conversion, standardisation, deduplication, relationship checks and JSON flattening.	Before-and-after values, survivor selection and exception records.
SCD Type 2	Change a tracked reservation attribute and verify that the old version expires and a new current version is inserted.	Non-overlapping effective dates and exactly one current row.
Gold layer	Confirm surrogate-key resolution, fact grain and aggregation logic.	Silver-to-Gold reconciliation and unresolved-key report.
KPI validation	Recalculate room revenue, ADR, recommendation rate and other measures using certified SQL.	Query output matching the Streamlit dashboard.
Security testing	Execute approved and restricted queries using each Snowflake role.	Successful authorised access and denied or masked unauthorised access.
Restart testing	Rerun a completed or partially failed batch.	No uncontrolled duplicate records and a traceable pipeline run ID.

20.1.1 Acceptance Criteria

- All 7,160 source records must have a documented processing outcome, and the 7,157 records approved by IDMC must be reproducible by dataset and validation rule.
- Every promoted reservation must reference valid guest, hotel and room records, except controlled late-arriving cases assigned to a monitored unknown member.

- Every reservation must have at most one current SCD Type 2 version, with no overlapping effective periods.
- Gold fact counts must reconcile to the eligible Silver change set after documented inserts, updates, duplicates and exclusions.
- Dashboard totals must match certified Snowflake SQL for the same property, reporting period, status and currency scope.
- The occupancy denominator and booking-funnel sequence must remain marked as under validation until reconciliation is complete.

20.2 Operational Monitoring and Deployment Readiness

Production readiness requires operational controls in addition to successful development. The platform should provide clear ownership, scheduling, monitoring, failure recovery and change-management procedures.

These controls ensure that hospitality managers receive timely information and that support teams can diagnose issues without manually reconstructing the complete pipeline.

Operational control	Recommended implementation	Business benefit
Pipeline scheduling	Define the execution sequence for S3 ingestion, IDMC mappings, Snowflake layer loads and Streamlit refresh.	Prevents incomplete data from appearing in the application.
Freshness monitoring	Record the latest successful batch and display the last refresh timestamp on every Streamlit page.	Allows users to judge whether reported information is current.
Volume monitoring	Compare current source and target counts with expected ranges and recent batch history.	Detects missing files, abnormal drops and duplicate deliveries.
Data-quality alerts	Raise alerts when critical rejects, nulls, duplicates or relationship failures exceed approved thresholds.	Prevents unreliable data from silently entering Gold marts.
Failure recovery	Document restart points, watermarks and idempotent MERGE behaviour.	Supports safe recovery without duplicate facts.
Security review	Review role grants, secure views and masking policies periodically.	Maintains least-privilege access to guest and payment data.
Release control	Version SQL, IDMC mappings and Streamlit code and record deployment approval.	Reduces uncontrolled changes and KPI drift.
Support ownership	Maintain a responsibility matrix for data engineering, revenue analytics and guest-experience issues.	Ensures incidents reach the correct owner quickly.

20.2.1 Deployment Readiness Checklist

- Complete source-to-target documentation for all seven datasets.
- Automate Bronze, Silver and Gold record-count reconciliation.
- Approve the KPI catalogue, including formula, grain, filters, owner and refresh schedule.
- Reconcile the available-room-night denominator used for occupancy.
- Validate event order, event duplication and session identity for booking conversion.
- Confirm that Streamlit queries only approved Gold views and not raw Bronze tables.
- Capture positive and negative role-based-access test evidence.
- Prepare batch restart, incident response and rollback procedures.
- Update the Kanban board so final documentation, PowerPoint and dashboard-enhancement status matches the delivered artefacts.

These controls provide a practical path from capstone demonstration to a stable analytical service.

The immediate priority is to close KPI reconciliation issues; the next priority is to automate monitoring and recovery; advanced Cortex and predictive use cases should be introduced only after the governed data foundation is consistently reliable.

21. Burn-down Chart

The burn-down chart compares planned remaining tasks with actual remaining tasks across the project period. The chart provides a concise measure of whether delivery is progressing at the expected rate and whether corrective action is required before the SME review.

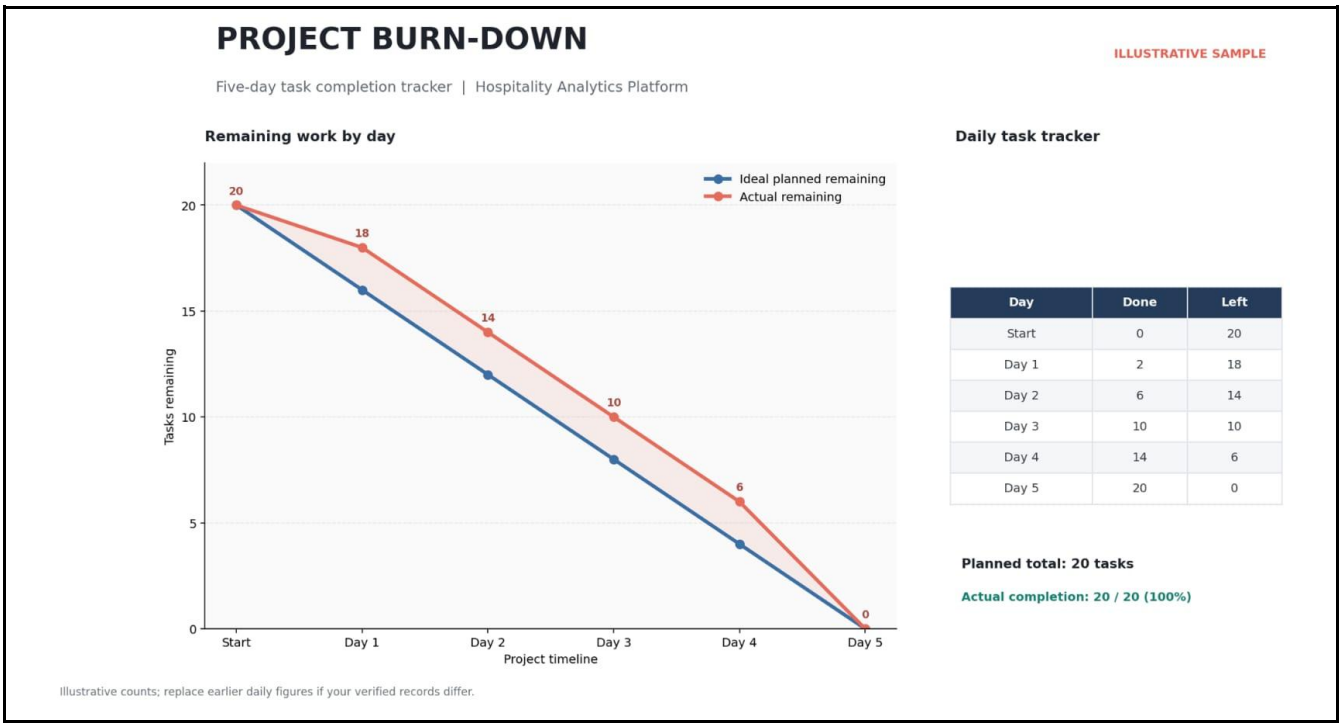


Figure 10. Project burn-down chart from the original presentation.

Interpretation

The project closed the planned task set within the delivery window. The actual line should be interpreted together with the Kanban board because a task count alone does not show the size or technical risk of each item. Any late addition, such as new Streamlit features, should be treated as scope change rather than hidden within the original baseline.

22. Kanban Chart

The Kanban board is the operational delivery view used by Capstone Project Team-06. The board separates backlog, in-progress and completed work and provides visible ownership for each work item.

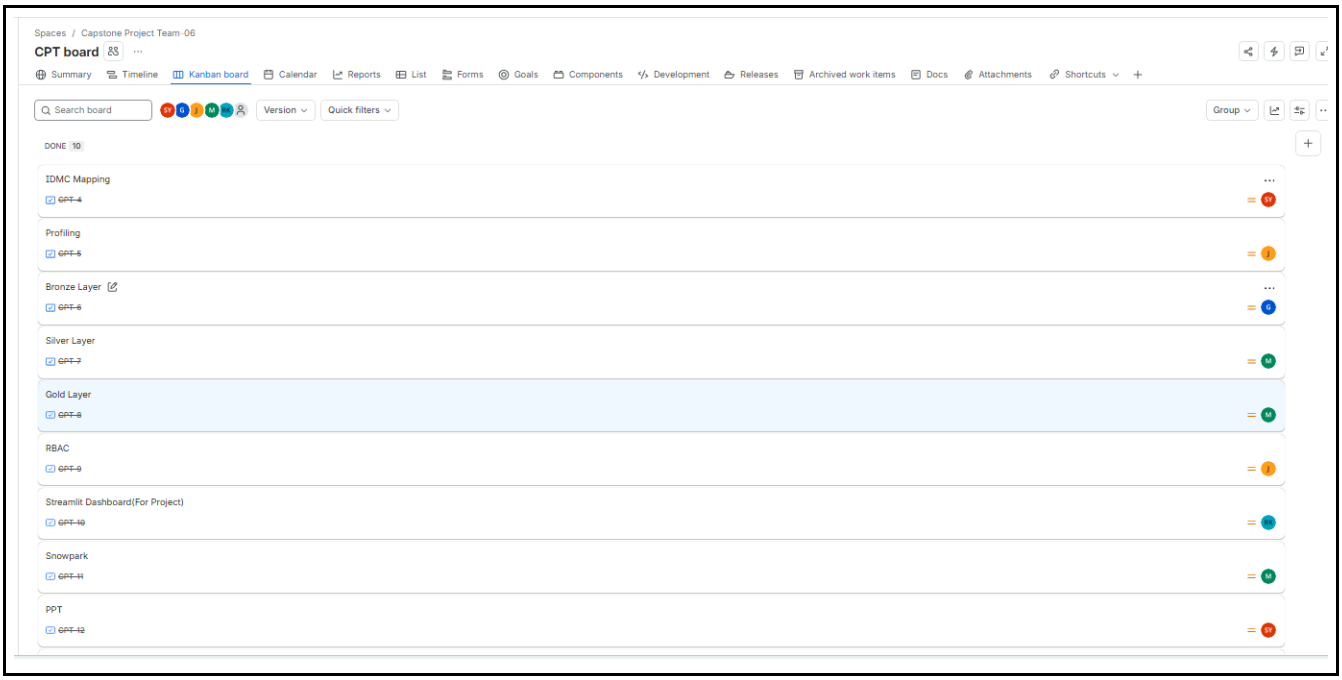


Figure 11. Actual Kanban board screenshot supplied by the project team.

Board status	Visible work items	Management interpretation
Backlog	PowerPoint and Documentation	Presentation and formal documentation remained open at the time of capture.
In Progress	Streamlit Dashboard for adding new features	The core dashboard existed and enhancement work was continuing.
Done	IDMC Mapping, Profiling, Bronze Layer, Silver Layer, Gold Layer, RBAC and Streamlit Dashboard for Project	The primary data pipeline, layered model, security foundation and first application version were completed.

The Kanban evidence should be read as a snapshot. Before final submission, the project team should update the board so that the status of documentation, PowerPoint preparation and additional Streamlit features matches the final delivered artefacts.

23. Conclusion and Future Enhancements

The project demonstrates a complete hospitality-data journey from source ingestion to business decision support. The strongest elements are the real IDMC implementation, explicit data-quality routing, Bronze-to-Silver-to-Gold separation, reservation-history processing, fact constellation model, role-based security, Streamlit application and combined guest-recovery and pricing USPs.

The report intentionally distinguishes confirmed implementation, rule-based analytics and future AI capability. Closing the occupancy, funnel and KPI reconciliation issues will improve management confidence and move the solution closer to production readiness.

Future enhancements

- Implement and validate a Cortex Analyst semantic model over approved Gold measures.
- Use Snowflake Cortex for governed sentiment classification and feedback summarisation.
- Create automated reconciliation and data-quality scorecards.
- Add pipeline monitoring, freshness alerts and restart runbooks.
- Track guest-recovery recommendations through assignment and closure.
- Develop predictive reservation-risk models only after labelled outcomes are available.
- Add near-real-time booking-event ingestion and robust sessionisation.
- Implement a central semantic KPI layer consumed by every Streamlit page.